

TANUSH THAKRAN

Software & Data · Python · SQL · React / Next.js · Machine Learning · Power BI
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PROFILE

Computer Science undergraduate working across software engineering, data and applied ML. Shipped production applications for a logistics client in Cairo, architected a multi-service SaaS backend on FastAPI and Celery, built a local RAG pipeline evaluated against a labelled question set, and delivered a SQL and Power BI profitability analysis that traced a company's margin problem to a specific pricing threshold. Comfortable owning a problem end to end — schema through deployment, or raw data through recommendation.

TECHNICAL SKILLS

Languages: Python · SQL · JavaScript (ES6+) · TypeScript · Java · Dart
Data & Analytics: PostgreSQL (window functions, CTEs) · Pandas · NumPy · scikit-learn · Power BI · DAX · Power Query · Google Analytics 4 · Excel · Cohort & RFM Analysis
Machine Learning: Retrieval-Augmented Generation · Embeddings & Vector Search · NLP & Text Processing · Whisper · Local LLMs (Ollama) · Retrieval Evaluation
Web & Mobile: React · Next.js 14 · Node.js · Express · FastAPI · Flutter · Tailwind CSS · REST API Design
Infrastructure: Docker & Docker Compose · Celery · Redis · MongoDB · Firebase · ChromaDB · Git · GitHub · Vercel · Render

PROJECTS

VideoCaptionMaker — Production SaaS Platform *Next.js 14 · FastAPI · Celery · PostgreSQL*
Architected a multi-service application across six Docker Compose services: Next.js 14 frontend, FastAPI backend, Celery workers, Redis, PostgreSQL and ChromaDB.
Built an async SQLAlchemy data layer with Alembic migrations and a typed API client; offloaded GPU-accelerated transcription to workers so the API stays responsive under load.
Designed security in from the first commit — httpOnly JWT cookies, bcrypt, CORS, upload validation — with auth built before the upload path existed.
FULL-STACK · DISTRIBUTED SYSTEMS · DOCKER · ASYNC PYTHON

Superstore Sales & Profitability Analysis *PostgreSQL · Power BI · DAX*
[github.com/Tanush1206/superstore-powerbi-analysis](#)
Analysed 9,994 transactions (\$2.3M revenue, 793 customers) to test whether a margin problem stemmed from product mix or pricing.
SQL bucketing showed margin holds at 29.5% undiscounted but turns negative above a 25% discount, reaching –77% past 40% — converting a vague concern into a specific policy threshold.
Validated the pattern independently at sub-category, region, category and customer level; wrote window-function and CTE queries for YoY growth, cohort retention and RFM segmentation.
Built an interactive Power BI dashboard with DAX measures and cross-filtering, delivering a recommendation worth an estimated \$35K in recoverable annual profit.
SQL · POWER BI · DAX · COHORT ANALYSIS · BUSINESS RECOMMENDATION

RAG-Based AI — Retrieval Pipeline & Evaluation *Python · Whisper · bge-m3 · Ollama*
[github.com/Tanush1206/rag_based_ai](#)
Built a local end-to-end pipeline: ffmpeg audio extraction, Whisper transcription, chunking, and vector representation via bge-m3 embeddings.
Implemented top-k cosine-similarity retrieval and tuned chunk size and overlap against a manually labelled question set, measuring improvement rather than assuming it.
Root-caused a crash that appeared only at scale by profiling embedding batch sizes, then restructured requests into chunked batches.
RAG · EMBEDDINGS · VECTOR SEARCH · EVALUATION

ATW Mobile — Driver & Medic Coordination App *Flutter · Firebase · Shipped for CYPARTA*
Built and shipped a production mobile app with live GPS tracking, deployed for a logistics client in Cairo.
Implemented full trip-lifecycle state management maintaining consistency across concurrent users on a real-time database.
FLUTTER · REAL-TIME SYSTEMS · PRODUCTION

EXPERIENCE

App Developer Intern — CYPARTA *Cairo, Egypt · Dec 2025 – Jan 2026*
Engineered post-trip EMS reporting modules with validation rules protecting data integrity across high-criticality logistics records.
Modelled full trip-lifecycle event data — request, acceptance, live status, completion — producing a clean, queryable operational record.
Worked with real-time GPS telemetry streams, handling ingestion, state updates and consistency across concurrent users.
PRODUCTION ENGINEERING · REAL-TIME DATA · DATA INTEGRITY

Junior Manager — AIESEC in Bengaluru *Oct 2025 – Mar 2026*
Tracked and reported departmental KPIs, using performance data to steer team priorities.
Managed the applicant pipeline end to end, monitoring stage-by-stage conversion and flagging drop-off points to leadership.
Led cross-functional coordination to hit targets and expand programme reach across the Bengaluru entity.
KPI REPORTING · FUNNEL ANALYSIS · LEADERSHIP

ADDITIONAL PROJECTS

PactPal — NLP contract analysis platform cutting measured reading time ~30% for non-expert users. *Python · NLP · React*
CLIQUE — Event tracking and centralisation platform for university students. *React · Node.js*
AASRAH — Humanitarian platform connecting users with NGOs, with Google Maps API location tracking. *React · Node/Express*

EDUCATION

BITS Pilani — Integrated B.Sc. in Computer Science *2024 – 2028*
Coursework: Algorithms, Data Structures, Database Systems, Operating Systems, Computer Networks, System Design, Statistics, Deep Learning.
Columbia Foundation School *2022 – 2024*
Class 11th & 12th — Science & Mathematics.

CERTIFICATIONS

Google Analytics Certification (GA4) — Google, Aug 2026 (valid to Aug 2027)
Intro to Machine Learning — Kaggle, Jul 2026

- [AI Fluency: Framework & Foundations — Anthropic, Jul 2026](#)
- [The Ultimate Job Ready Data Science Course — CodeWithHarry](#)